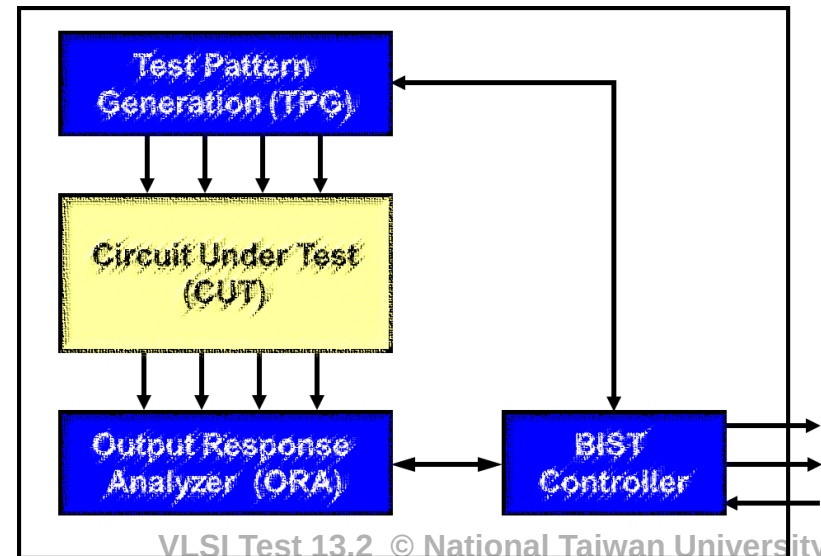


BIST Part1 - TPG

- Introduction
- Test Pattern Generation (TPG)
 - ◆ Deterministic: ROM Algorithm Counter
 - ◆ Pseudo Random
 - * Linear Feedback Shift Register, LFSR (1977)
 - Two types of LFSR
 - Design of LFSR
 - * Cellular Automata, CA (1984)

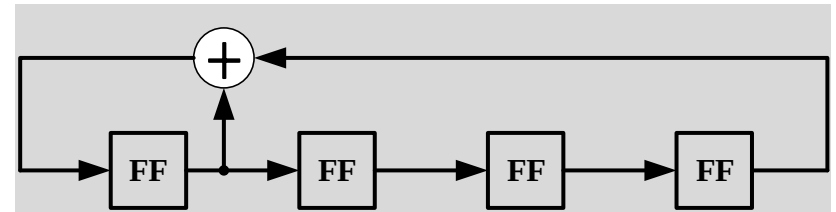


Linear Feedback Shift Register [Frohwerk 77]

- LFSR consist of unit delays (flip-flops, **FF**) and feedback (**XOR**)
- Two applications of LFSR:

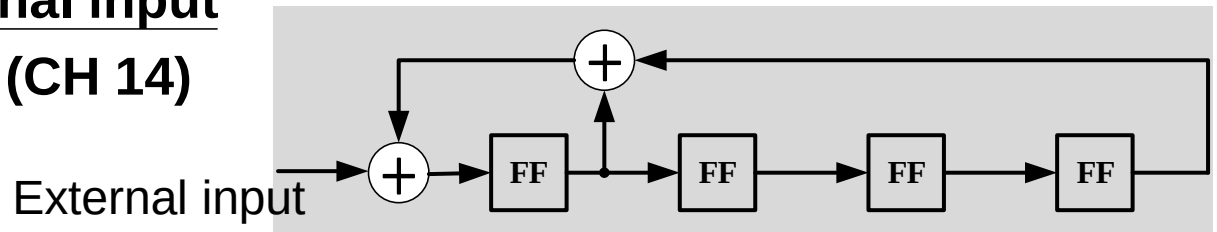
1. LFSR without external input

- * Used for **TPG**
- * aka **Autonomous LFSR**



2. LFSR with external input

- * Used for **ORA** (CH 14)



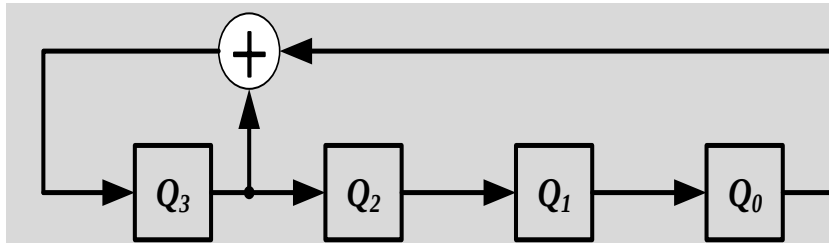
$\oplus$ = XOR

- “Linear” because XOR is **mod-2 addition**

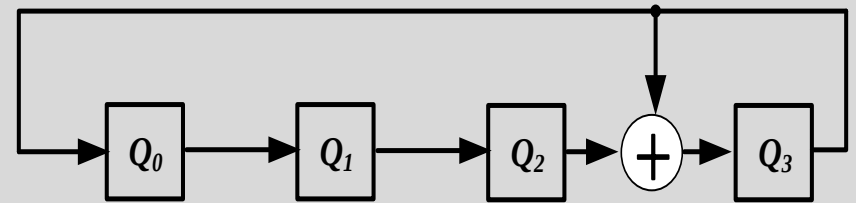
Two Types of Autonomous LFSR

- Autonomous LFSR is **Modular Counter**
 - ♦ Very **small area**, generate **pseudo random outputs**
- Two structures:
 - ♦ **Type 1: Standard Form (aka external XOR) LFSR**
 - ♦ **Type 2: Modular Form (aka internal XOR) LFSR**

Type-1 Standard LFSR



Type-2 Modular LFSR



LFSR is Good for TPG

Type-1, Standard Form LFSR

- Three ways to describe LFSR

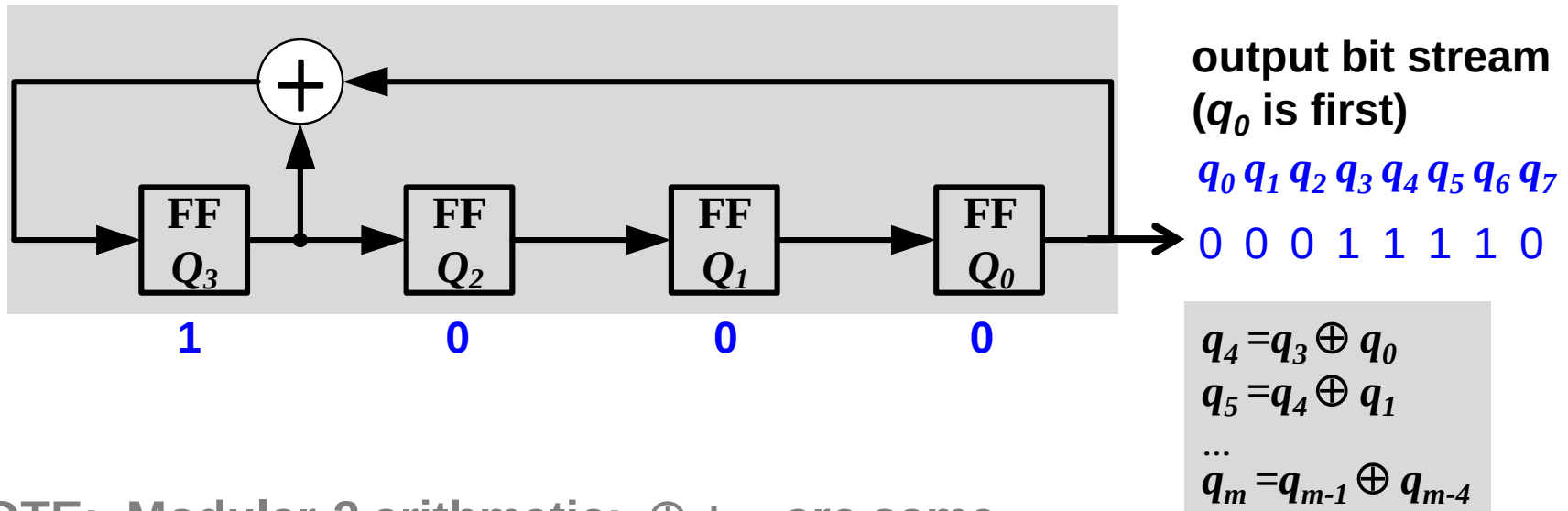
1. Next state equation: $Q_3^+ = Q_3 \oplus Q_0$

* Q_3^+ means next state of FF Q_3

2. Recurrence equation: $q_m = q_{m-1} \oplus q_{m-4}$

3. Characteristic polynomial: $f(x) = x^4 + x^3 + 1$

* Most popular. will see why 13.4

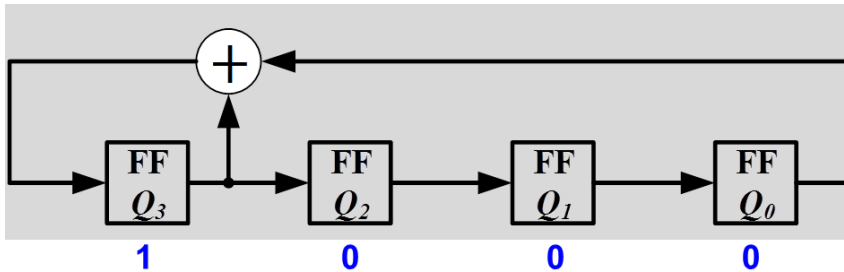


- NOTE: Modular-2 arithmetic: $\oplus$ + - are same

- ♦ $1 \oplus 1 = 0$; $0 \oplus 1 = 1$; $1 \oplus 0 = 1$; $0 \oplus 0 = 0$

State Sequence of x^4+x^3+1 LFSR

- **Seed** = Initial state of LFSR
 - ♦ must be non-zero
- Total $2^4-1 = 15$ distinct states
 - ♦ All-zero state not included
- **Periodical.**
 - ♦ Cycle length $L_c = 15$

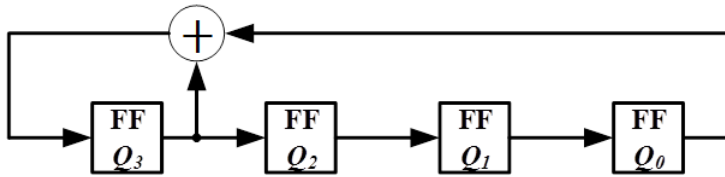


Back to seed after 15 cycles →

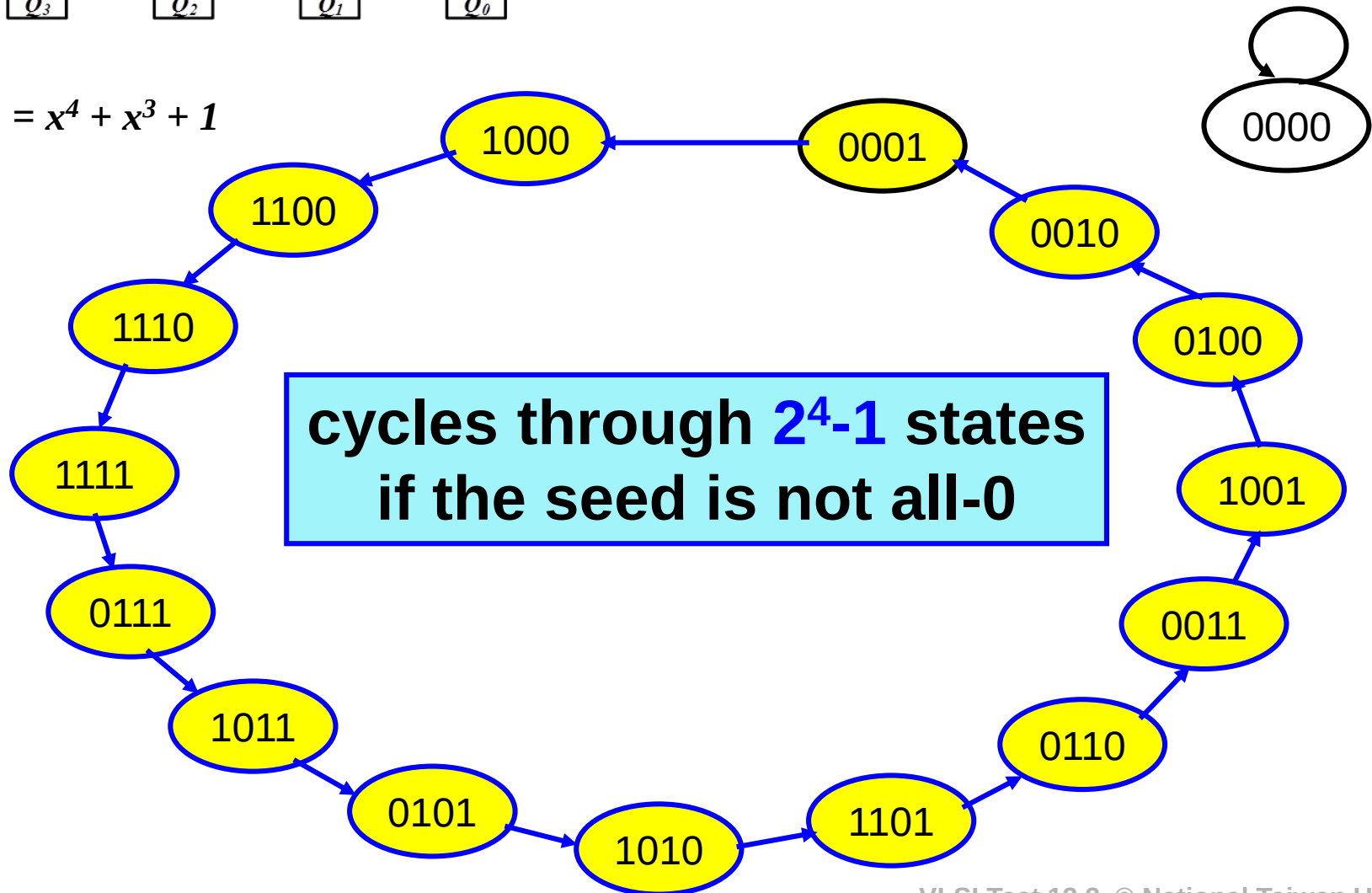
state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 (=0)	1	0	0	0

seed

State Diagram

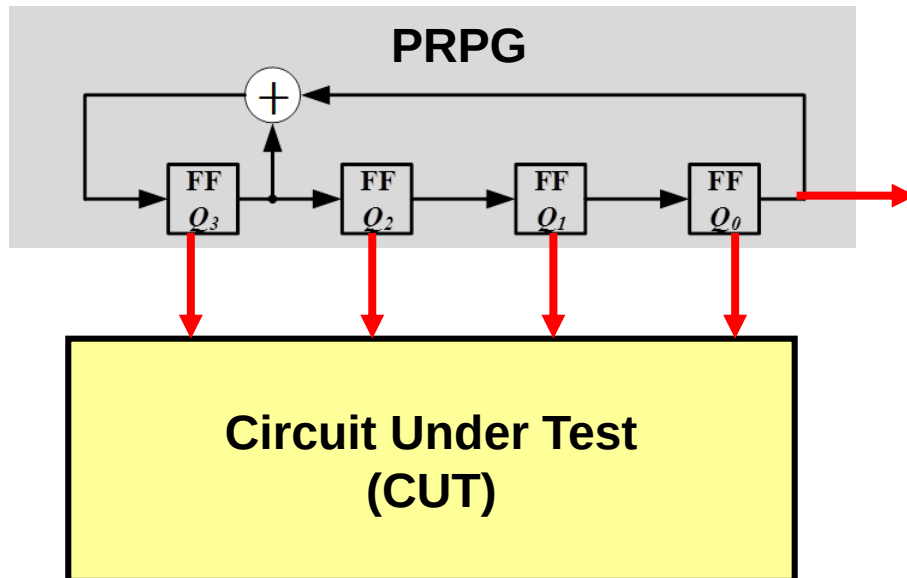


$$f(x) = x^4 + x^3 + 1$$



Pseudo Random Pattern Generator (PRPG)

- **Pseudo random.** NOT truly random
- Serial PRPG: $q_0, q_1, q_2, q_3, \dots$
 - ♦ Periodical. $L_c = 15$
- Parallel PRPG: ($Q_3 \ Q_2 \ Q_1 \ Q_0$)
 - ♦ Each output **shifted by one cycle**
 - ♦ **Phase difference** = 1



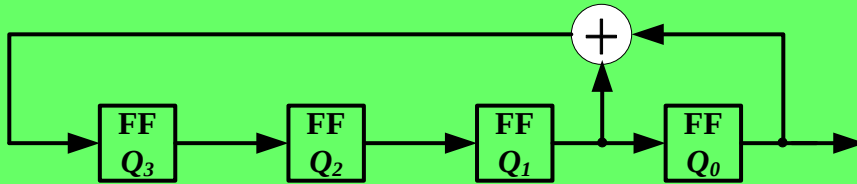
state	Q_3	Q_2	Q_1	Q_0	
0	1	0	0	0	q_0
1	1	1	0	0	q_1
2	1	1	1	0	q_2
3	1	1	1	1	q_3
4	0	1	1	1	q_4
5	1	0	1	1	q_5
6	0	1	0	1	q_6
7	1	0	1	0	q_7
8	1	1	0	1	q_8
9	0	1	1	0	q_9
10	0	0	1	1	q_{10}
11	1	0	0	1	q_{11}
12	0	1	0	0	q_{12}
13	0	0	1	0	q_{13}
14	0	0	0	1	q_{14}
15 (=0)	1	0	0	0	q_{15}

↔ ↔ ↔

Quiz

Q: Given this new LFSR, show the state sequences with seed [1 0 0 0]

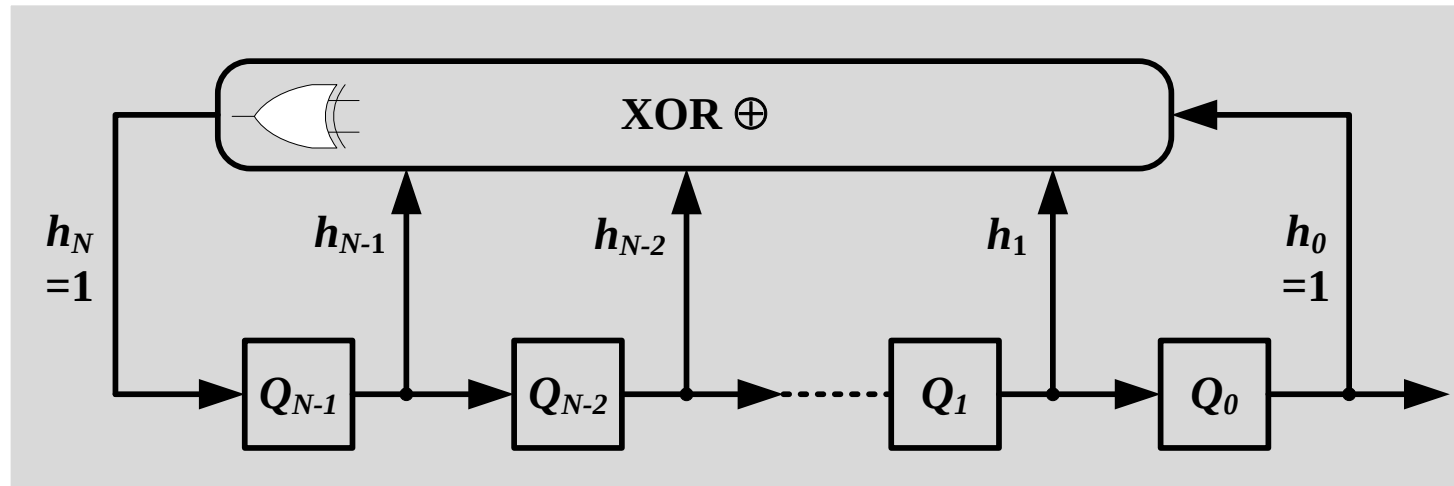
Q2: What is cycle length L_c = ?



state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

N-degree Standard Form LFSR

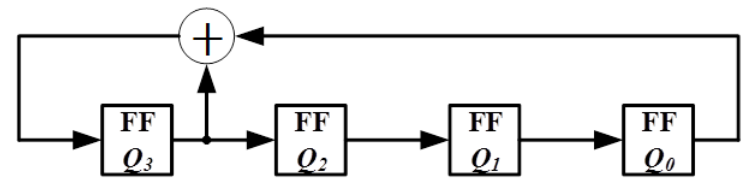
- $N = \text{LFSR degree} = \text{number of FF} = \text{characteristic polynomial degree}$



$h_i = 1$ if feedback exists

$h_i = 0$ if no feedback

$h_N = 1, h_0 = 1$ 



$$f(x) = x^4 + x^3 + 1$$

- **Characteristic polynomial:** (details see appendix)

$$f(x) = \sum_{i=0}^N h_i x^i$$

$f(x)$ Can Be Written from LFSR Structure

Matrix Representation (Type-1 LFSR)

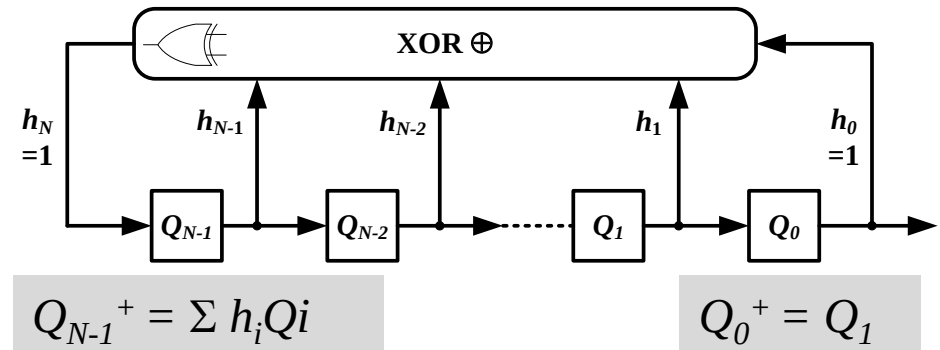
$$\begin{bmatrix} Q_0^+ \\ Q_1^+ \\ \vdots \\ \vdots \\ Q_{N-3}^+ \\ Q_{N-2}^+ \\ Q_{N-1}^+ \end{bmatrix} = \begin{bmatrix} \text{Identity matrix} \\ h_0 \quad h_1 \quad h_2 \quad \dots \quad h_{N-2} \quad h_{N-1} \end{bmatrix} \begin{bmatrix} Q_0 \\ Q_1 \\ \vdots \\ \vdots \\ Q_{N-3} \\ Q_{N-2} \\ Q_{N-1} \end{bmatrix}$$

$$Q^+ = T Q \pmod{2}$$

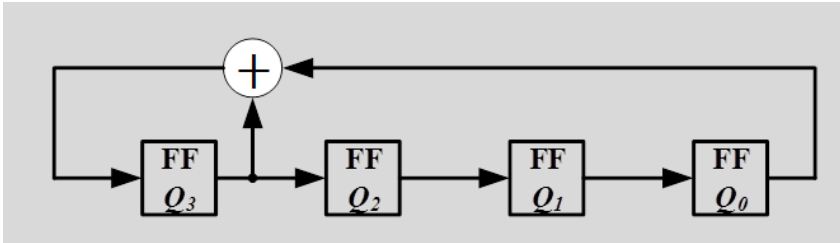
Q = current state

Q^+ = next state

T = **companion matrix**



Example



state	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1

$$T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$T^2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

mod-2 arithmetic:
 $1 \times 1 = 1$ $1 \times 0 = 0$ $0 \times 0 = 0$
 $1 + 1 = 0$ $1 + 0 = 1$ $0 + 0 = 0$

$$Q_{seed} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$TQ_{seed} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

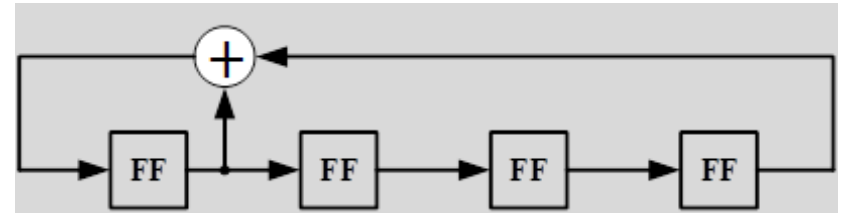
$$T^2Q_{seed} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$T^3Q_{seed} = T^2TQ_{seed} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

**Matrix Multiplication
Same as Simulation**

Linear Relationship between States

- 0th cycle: LFSR state = Q_{seed}
- 1st cycle: TQ_{seed}
- 2nd cycle: $TTQ_{seed} = T^2Q_{seed}$
- 4th cycle: $T^4Q_{seed} = (T^2)^2Q_{seed}$
- 16th cycle: $T^{16}Q_{seed} = (T^4)^2Q_{seed} = ((T^2)^2)^2Q_{seed}$
 - ♦ only 3 matrix squares needed
- After L_c cycles, LFSR returns to initial state
 - ♦ $T^{L_c}Q_{seed} = Q_{seed}$
 - ♦ In this case $L_c=15$. $T^{15} = I$, so $T^{16} = T$



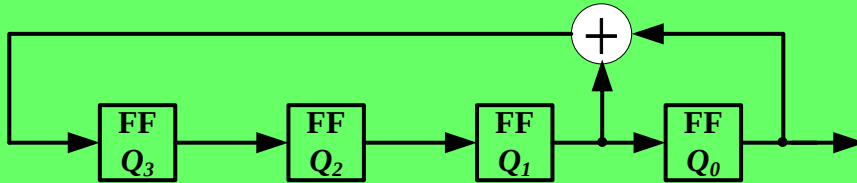
Matrix Faster Than Simulation

Quiz

Q1: Given this LFSR, What is its characteristic polynomial?

Q2: What is its companion matrix T ?

Q3: Use T to derive state after three cycles, starting from seed $[1\ 0\ 0\ 0]$



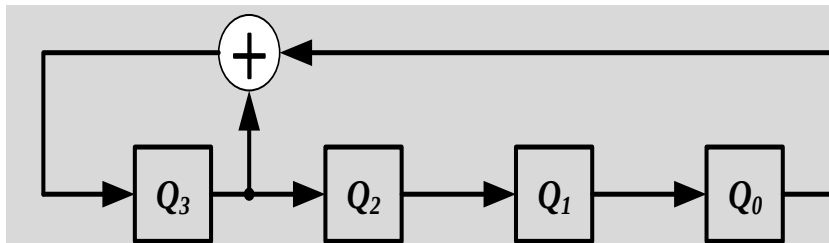
state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	1	0	0	1

ANS

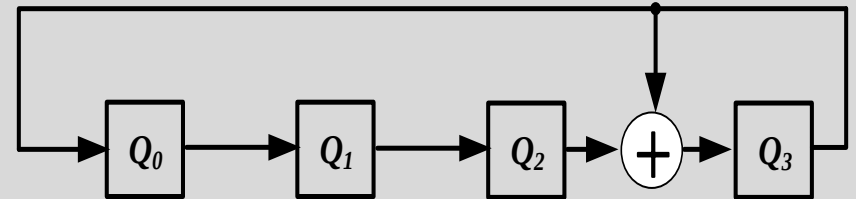
LFSR Types

- **Type 1: Standard Form (aka external XOR) LFSR**
- **Type 2: Modular Form (aka internal XOR) LFSR**
 - ♦ XOR gates are **internal** to LFSR
 - ♦ as opposed to standard form LFSR, in which XOR are **external**

Type-1 Standard LFSR



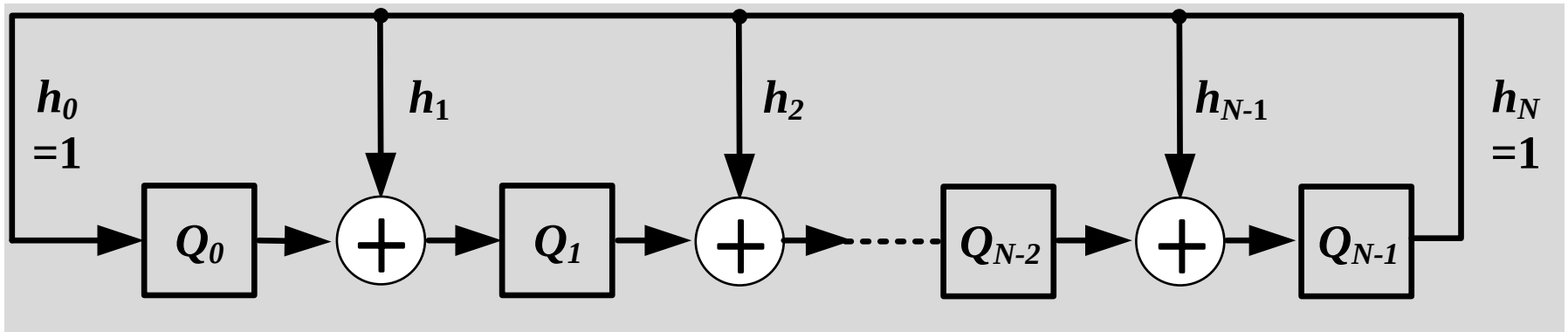
Type-2 Modular LFSR



NOTE: Order of FF Different

N-degree Modular Form LFSR

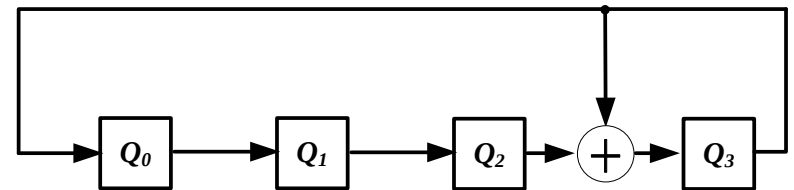
- **$N = \text{LFSR degree}$** = number of FF = characteristic polynomial degree



$h_i = 1$ if feedback exists

$h_i = 0$ if no feedback

$h_N = 1, h_0 = 1$ 



$$f(x) = x^4 + x^3 + 1$$

- **Characteristic polynomial:**

$$f(x) = \sum_{i=0}^N h_i x^i$$

$f(x)$ Same as Type-1 LFSR
(remember correct order of FF !)

Matrix Representation (Type-2 LFSR)

$$\begin{bmatrix} Q_0^+ \\ Q_1^+ \\ \vdots \\ \vdots \\ Q_{N-3}^+ \\ Q_{N-2}^+ \\ Q_{N-1}^+ \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ \vdots & 1 & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 1 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ h_2 \\ \vdots \\ h_{N-2} \\ h_{N-1} \end{bmatrix} \begin{bmatrix} Q_0 \\ Q_1 \\ \vdots \\ \vdots \\ Q_{N-3} \\ Q_{N-2} \\ Q_{N-1} \end{bmatrix}$$

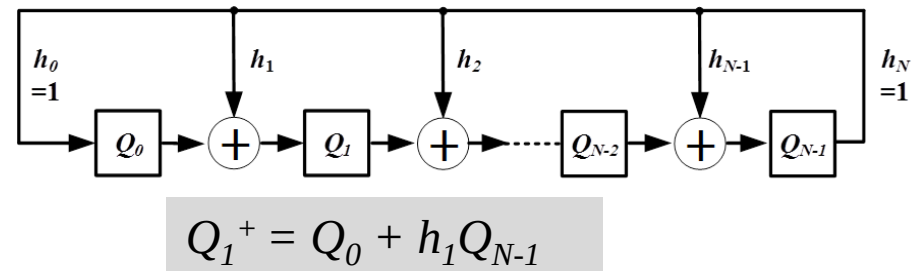
Identity matrix

$$Q^+ = T Q$$

Q = current state

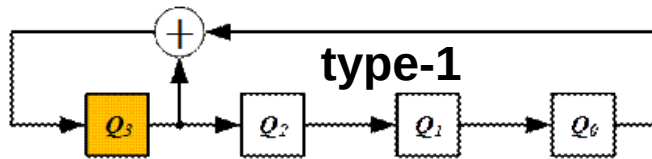
Q^+ = next state

T = Companion matrix

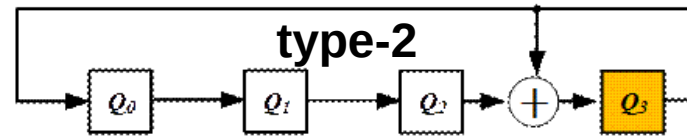


Quiz

Q: Given two types LFSR of x^4+x^3+1 , same seed 1000, fill in table.



state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 (=0)	1	0	0	0



state	Q_0	Q_1	Q_2	Q_3
0	1	0	0	0
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15 (=0)				

Summary – LFSR

- Two types of LFSR. Simple structure, good for TPG
- LFSR generate pseudo random test patterns. Repeat every L_c cycles
- Linear relationship between states: $Q^+=TQ$
- Characteristic polynomial is often used to describe LFSR

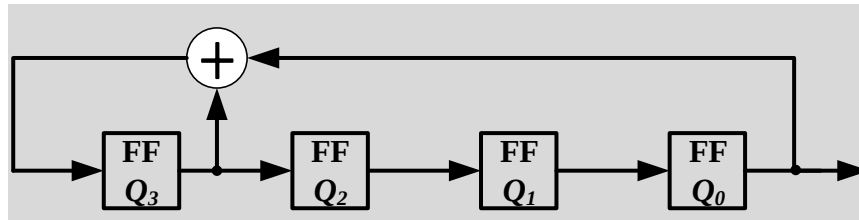
	Structure	Matrix
type1 standard-form	index right to left	$\begin{bmatrix} Q_0^+ \\ Q_1^+ \\ \vdots \\ Q_{N-3}^+ \\ Q_{N-2}^+ \\ Q_{N-1}^+ \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \\ h_0 & h_1 & h_2 & \dots & h_{N-2} & h_{N-1} \end{bmatrix} \begin{bmatrix} Q_0 \\ Q_1 \\ \vdots \\ Q_{N-3} \\ Q_{N-2} \\ Q_{N-1} \end{bmatrix}$
type2 modular-form	index left to right	$\begin{bmatrix} Q_0^+ \\ Q_1^+ \\ \vdots \\ Q_{N-3}^+ \\ Q_{N-2}^+ \\ Q_{N-1}^+ \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & h_0 \\ 1 & 0 & 0 & \dots & 0 & h_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 1 & \dots & 0 & \vdots \\ 0 & 0 & 0 & \dots & 0 & h_{N-2} \\ 0 & 0 & 0 & \dots & 1 & h_{N-1} \end{bmatrix} \begin{bmatrix} Q_0 \\ Q_1 \\ \vdots \\ Q_{N-3} \\ Q_{N-2} \\ Q_{N-1} \end{bmatrix}$
Char. Poly	$f(x) = \sum_{i=0}^N h_i x^i$	$f(\lambda) = \det(T - \lambda I)$

FFT 1

- Characteristic polynomial **can also be derived from matrix T**

$$f(\lambda) = \det(T - \lambda I)$$

- Q1: Given this type-1 LFSR, Please verify that $\det(T - \lambda I) = \lambda^4 - \lambda^3 - 1$
- Q2: From structure, we know $= f(x) = x^4 + x^3 + 1$
 - Not the same, why?

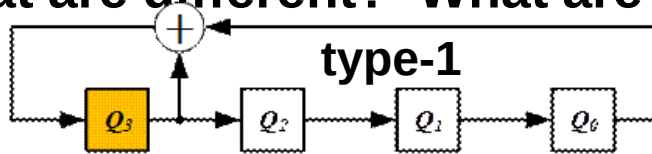


$$T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

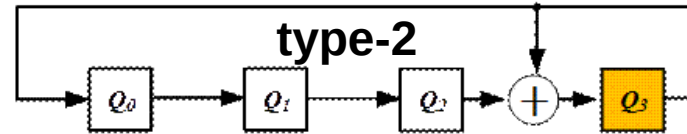
$$\det(T - \lambda I) = \begin{vmatrix} -\lambda & 1 & 0 & 0 \\ 0 & -\lambda & 1 & 0 \\ 0 & 0 & -\lambda & 1 \\ 1 & 0 & 0 & 1-\lambda \end{vmatrix} = ?$$

FFT 2

- Q: Given two types LFSR of same x^4+x^3+1 , same seed 1000.
- What are different? What are same?

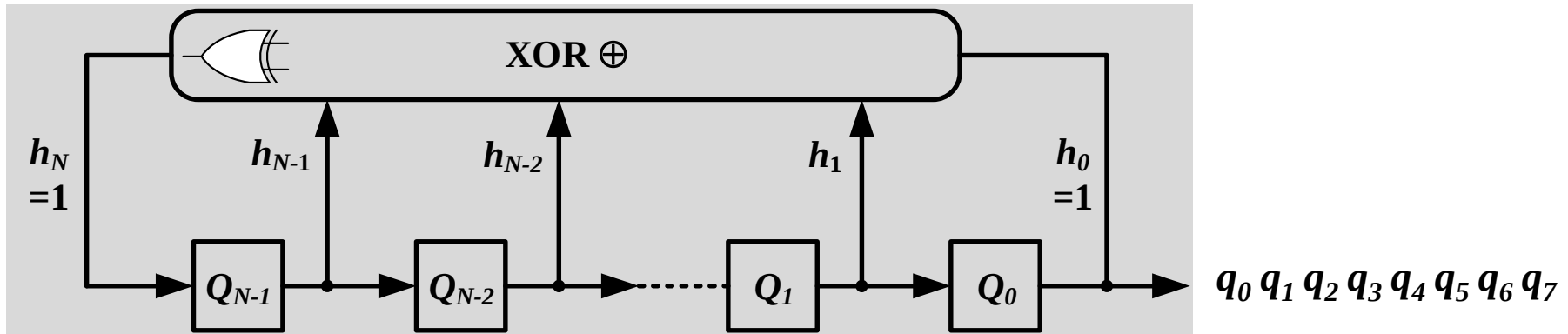


state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 (=0)	1	0	0	0



state	Q_0	Q_1	Q_2	Q_3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	0	1
5	1	1	0	1
6	1	1	1	1
7	1	1	1	0
8	0	1	1	1
9	1	0	1	0
10	0	1	0	1
11	1	0	1	1
12	1	1	0	0
13	0	1	1	0
14	0	0	1	1
15 (=0)	1	0	0	0

APPENDIX: Characteristic Polynomial (1)



output bit stream: $q_0, q_1, q_2, \dots$ (q_0 first)

recurrence equation:
$$q_m = \sum_{i=1}^N h_i q_{m-i}$$

generating function:
$$G(x) = q_0 x^0 + q_1 x^1 + q_2 x^2 + \dots = \sum_{m=0}^{\infty} q_m x^m$$

$$= \sum_{m=0}^{\infty} \left[\left(\sum_{i=1}^N h_i q_{m-i} \right) \cdot x^m \right] = \sum_{i=1}^N h_i x^i \cdot \left[\sum_{m=0}^{\infty} q_{m-i} x^{m-i} \right]$$

$$= \sum_{i=1}^N h_i x^i \cdot [q_{-i} x^{-i} + \dots + q_{-1} x^{-1} + \sum_{m=0}^{\infty} q_m x^m]$$

APPENDIX: Characteristic Polynomial (2)

$$G(x) = \sum_{i=1}^N h_i x^i \cdot [q_{-i} x^{-i} + \dots + q_{-1} x^{-1} + G(x)]$$

$$G(x) = \frac{\sum_{i=1}^N h_i x^i \cdot [q_{-i} x^{-i} + \dots + q_{-1} x^{-1}]}{1 - \sum_{i=1}^N h_i x^i}$$

 **LFSR seed. $q_{-N}, \dots, q_{-1}$**
 **LFSR structure**

when seed $q_{-N} \dots q_{-1} = [1000 \dots 0]$

$$G(x) = \frac{1}{1 - \sum_{i=1}^N h_i x^i} = \frac{1}{f(x)}$$

characteristic polynomial: $f(x) = 1 - \sum_{i=1}^N h_i x^i = \sum_{i=0}^N h_i x^i$